



IDBI INN@VATE

2026

Build. Integrate. Transform.

Knowledge Partner



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Team Details

- a. **Team name:** Black Bulls
- b. **Team leader name:** Shubham
- c. **Problem Statement:** Open Track — AI copilot for channel-sourced retail lending (conversational AI · product advisory · financial-health scoring · default prediction)

Brief about the idea

THE IDEA

A channel-aware loan-origination & partner-relationship copilot — on top of IDBI's LOS / BRE / LMS

Retail lending is sourced through DSAs & dealers and runs on TAT + trust. We work all four sides at once:



Customer

Only suitable, clearly-explained, instant offers — no spam, no black box.



RM / Bank

A queue ranked by risk-adjusted profit, auto-scoring & recommendations.



Channel partner (DSA)

Instant decisions, transparent payouts, tiering — the loyalty moat.



Risk / Regulator

Explainable, fair, adverse-selection-aware lending with audit trail.

Lend faster, safer, and win partner loyalty.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it's different

- ◆ Channel-first, not customer-only — models the DSA relationship: adverse-selection, TAT, tiering, churn.
- ₹ Bank-grade economics — ranks by $P(\text{convert}) \times (\text{net interest income} - \text{expected credit loss})$.
- ▲ Explainable — per-decision reason codes + Reg-B / RBI adverse-action codes; fair-lending exclusions.
- Integrates, not replaces — an intelligence layer on top of LOS / BRE / LMS.

How it solves the problem

- ✓ Right lead → right product → right price, instantly, with a professional offer.
- ✓ Flags risky sourcing before disbursal → fewer NPAs.
- ✓ Transparent, fast, tiered payouts → partner loyalty → more & better volume.
- ★ The flywheel: win the partner → volume → data → sharper models.

List of features offered by the solution

FEATURES

Customer & RM

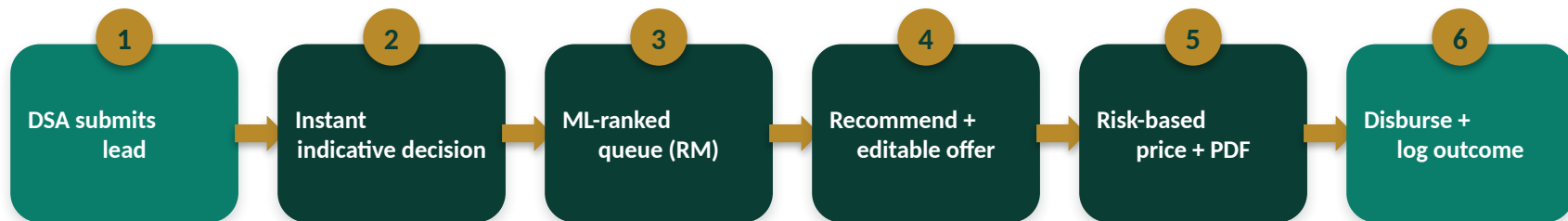
- ✓ ML conversion-propensity queue (risk-adjusted)
- ✓ Explainable financial-health score (0-100)
- ✓ Deterministic eligibility, rate & EMI engine
- ✓ Suitability-first next-best-product advisory
- ✓ Editable offer + risk-based pricing in-band
- ✓ Premium branded PDF + WhatsApp / Gmail
- ✓ Per-decision reason codes on every score

Channel & Partner

- ✓ Partner/lead default-risk (adverse-selection)
- ✓ Channel-mix ROI + book concentration (HHI)
- ✓ Commission + clawback payout engine
- ✓ Duplicate / fraud lead detection
- ✓ Instant decision (<50 ms) + SLA / TAT clock
- ✓ Partner tiering + churn early-warning
- ✓ Agentic AI assistant (natural language)

Process flow diagram or Use-case diagram

PROCESS FLOW



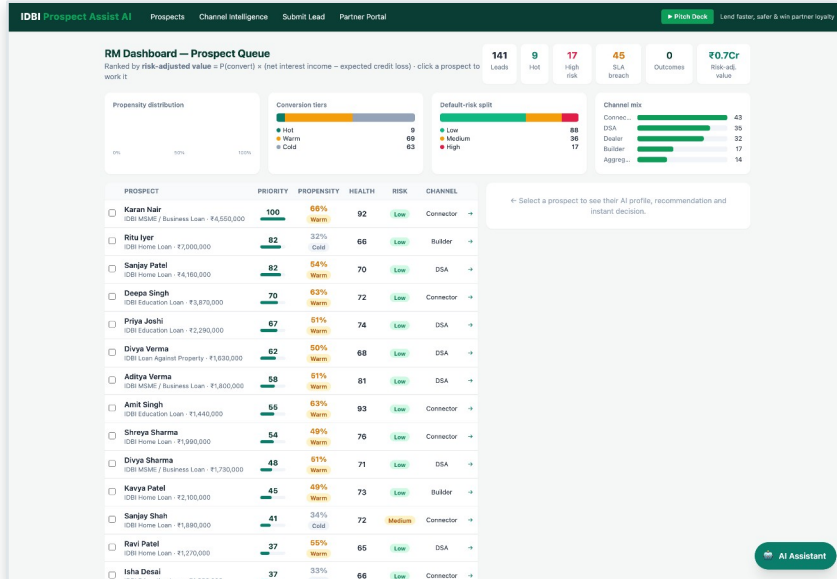
🔄 Outcomes (won / lost / default) retrain the ML models — the compounding data moat

Numbers boundary: the deterministic engine owns every rupee (eligibility, rate, EMI, ECL); ML owns probabilities; the LLM only phrases. RM stays in control — edits re-validate through the engine.

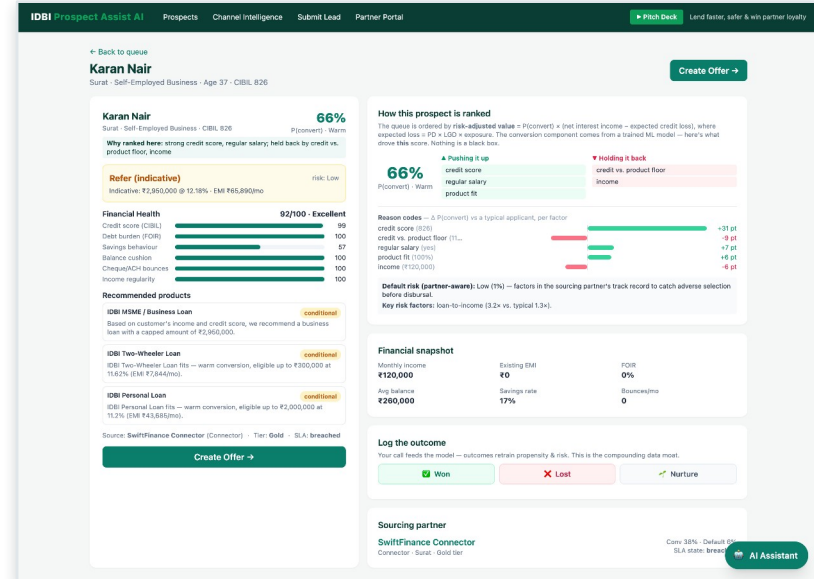
Wireframes/Mock diagrams of the proposed solution (optional)

INTERFACE

RM dashboard — risk-adjusted, ML-ranked queue

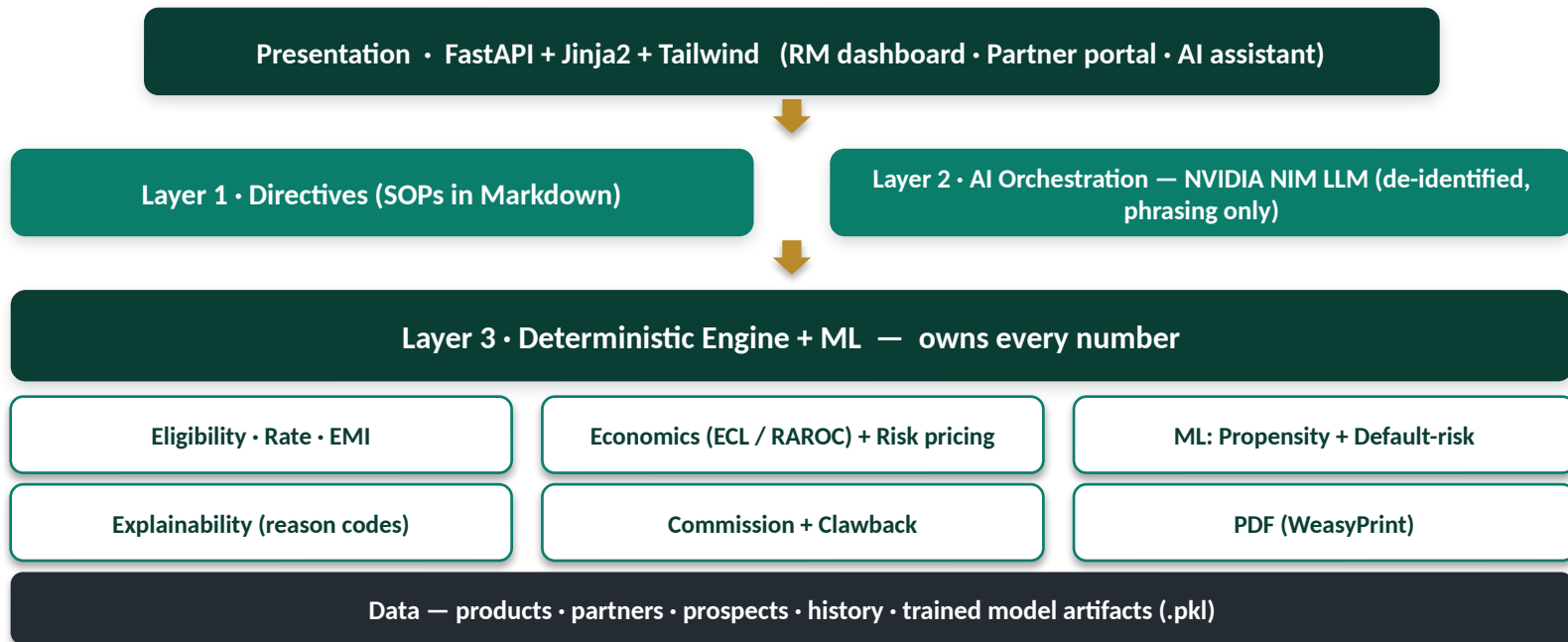


Prospect profile — reason codes & recommendation



Architecture diagram of the proposed solution

ARCHITECTURE



Technologies to be used in the solution

TECHNOLOGY



Backend & Web

Python 3.14

FastAPI

Uvicorn

Jinja2

Tailwind CSS

vanilla JS



Machine learning & AI

scikit-learn

NumPy / SciPy

joblib

occlusion XAI

NVIDIA NIM LLM



Documents & PDF

WeasyPrint

pango / cairo



Quality & Deployment

pytest · 44 tests

Playwright

Docker

Render / Railway / Cloud Run

Estimated implementation cost (optional)

INVESTMENT

PROTOTYPE — TODAY

≈ ₹0

Open-source stack
NVIDIA NIM free tier
Free-tier hosting · no licence cost

PRODUCTION PoC — INDICATIVE (ANNUAL)

- ▶ LLM / GPU inference — usage-based
- ▶ Cloud hosting + DB + storage — modest
- ▶ Integration — LOS/BRE/KYC/CIBIL (one-time)
- ▶ Model-ops — monitoring, retraining, governance

ROI: cost saved (NPA reduction + payout-leakage → 0 + RM selling hours) » cost to run

Snapshots of the prototype

PROTOTYPE

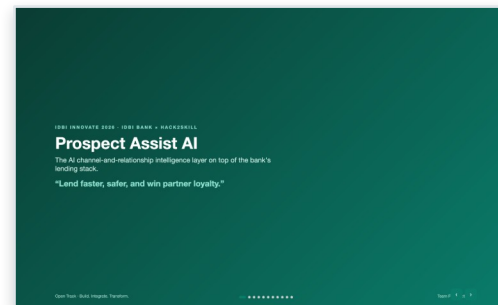
Editable offer — live EMI & risk-based pricing

Instant decision + adverse-action reason codes

Channel intelligence — adverse selection, ROI, HHI

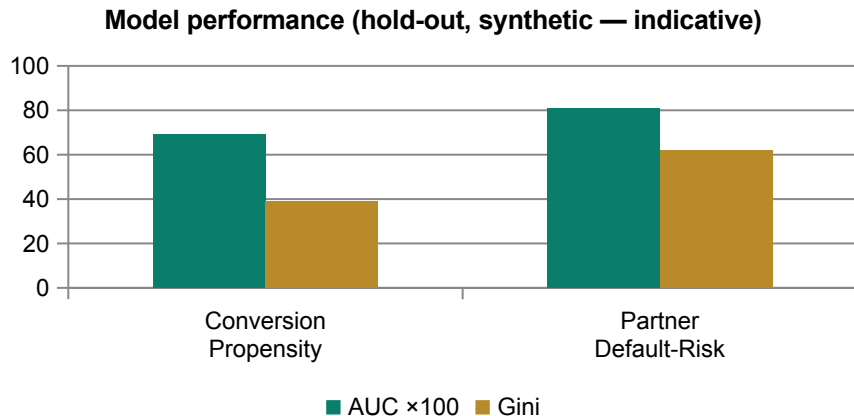
Partner	Product	ROI	HHI	Adverse Selection
Axis Bank	Axis Bank	87	45%	3%
Axis Bank	Axis Bank	85	55%	7%
Axis Bank	Axis Bank	80	55%	8%
Axis Bank	Axis Bank	78	55%	7%
Axis Bank	Axis Bank	78	60%	7%
Axis Bank	Axis Bank	77	65%	8%
Axis Bank	Axis Bank	76	55%	10%
Axis Bank	Axis Bank	75	65%	10%
Axis Bank	Axis Bank	75	65%	5%
Axis Bank	Axis Bank	75	30%	5%
Axis Bank	Axis Bank	75	45%	8%
Axis Bank	Axis Bank	75	45%	8%
Axis Bank	Axis Bank	75	45%	8%

Built-in pitch deck (live numbers)



Prototype Performance report/Benchmarking

BENCHMARKS



< 50 ms

Instant decision latency

~1–2 s

LLM reply · 9 s timeout

0.22

Partner-history feature share — no leakage · Bayesian shrinkage

Business impact on the synthetic book

₹16.4 Cr

pipeline value

₹0.68 Cr

risk-adjusted value

44 / 44

automated tests passing

Additional Details/Future Development (if any)

ROADMAP

1 Phase 1 • Integrate

- ▶ IDBI sandbox APIs
- ▶ CIBIL bureau + KYC
- ▶ e-consent (DPDP Act)
- ▶ LOS / BRE / LMS hooks

2 Phase 2 • Harden

- ▶ Retrain on real conv / NPA / TAT
- ▶ SHAP + PSI drift monitoring
- ▶ Auth / RBAC + audit trail
- ▶ Data residency

3 Phase 3 • Scale

- ▶ Partner mobile portal + WhatsApp
- ▶ OTP-gated secure delivery
- ▶ Collections & Early-Warning
- ▶ NVIDIA AI Enterprise / on-prem

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link



GitHub Public Repository

<https://github.com/Shubham-33/Hack2Skill-IDBI-Hackathone>



Final Product Link (live prototype)

<https://prospect-assist-ai-2guf.onrender.com>



Demo Video Link (3 minutes)

[paste your unlisted YouTube / Drive link]



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Thank you!